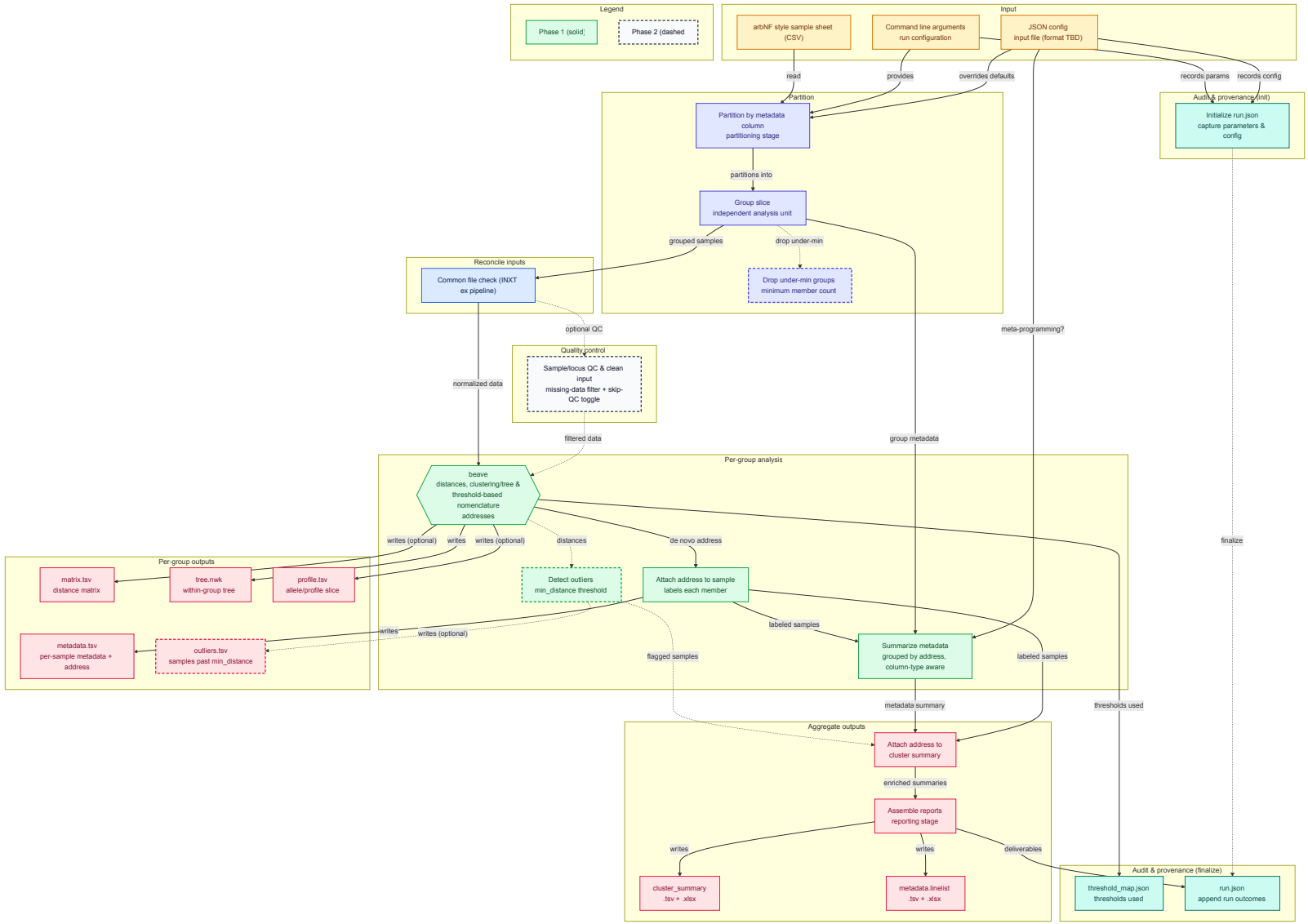

Legend
Phase 1 (solid)
Phase 2 (dashed
Input
arbNF style sample sheet (CSV)
Command line arguments run configuration
JSON config input file (format TBD)
read
provides
overrides defaults
records params
records config
Partition
Partition by metadata column partitioning stage
partitions into
Group slice independent analysis unit
grouped samples
drop under-min
Drop under-min groups minimum member count
Audit & provenance (init)
Initialize run.json capture parameters & config
Reconcile inputs
Common file check (INXT ex pipeline)
optional QC
meta-programming?
Quality control
Sample/locus QC & clean input missing-data filter + skip-QC toggle
normalized data
filtered data
group metadata
Per-group analysis
beave distances, clustering/tree & threshold-based nomenclature addresses
writes (optional)
writes
writes (optional)
distances
de novo address
finalize
Per-group outputs
matrix.tsv distance matrix
tree.nwk within-group tree
profile.tsv allele/profile slice
Detect outliers min_distance threshold
Attach address to sample labels each member
metadata.tsv per-sample metadata + address
writes
writes (optional)
outliers.tsv samples past min_distance
labeled samples
flagged samples
Summarize metadata grouped by address, column-type aware
labeled samples
metadata summary
thresholds used
Aggregate outputs
Attach address to cluster summary
enriched summaries
Assemble reports reporting stage
writes
writes
deliverables
cluster_summary .tsv + .xlsx
metadata.linelist .tsv + .xlsx
Audit & provenance (finalize)
threshold_map.json thresholds used
run.json append run outcomes